National College of Ireland



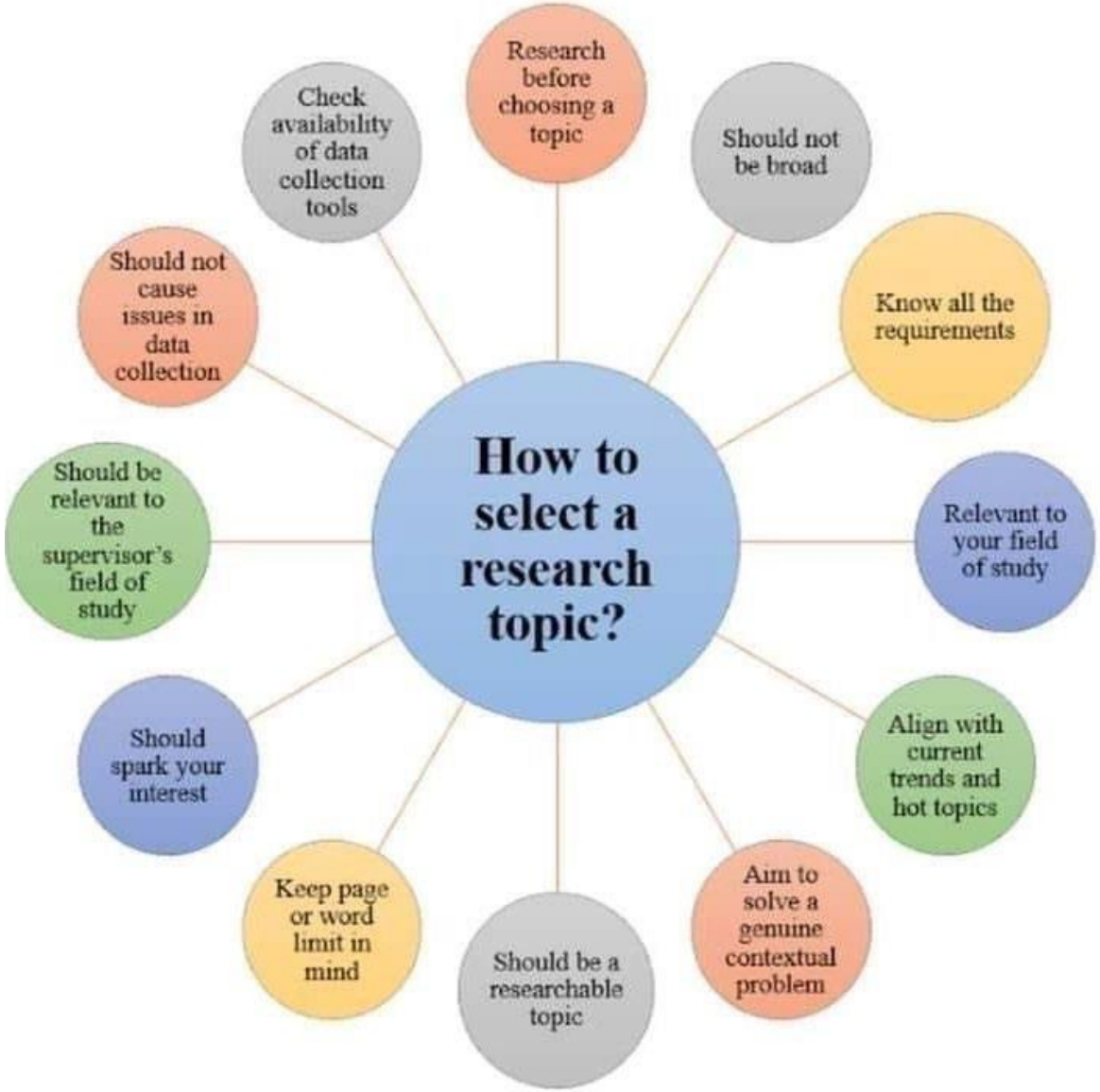
Research In Computing

Week 1&2: Lab

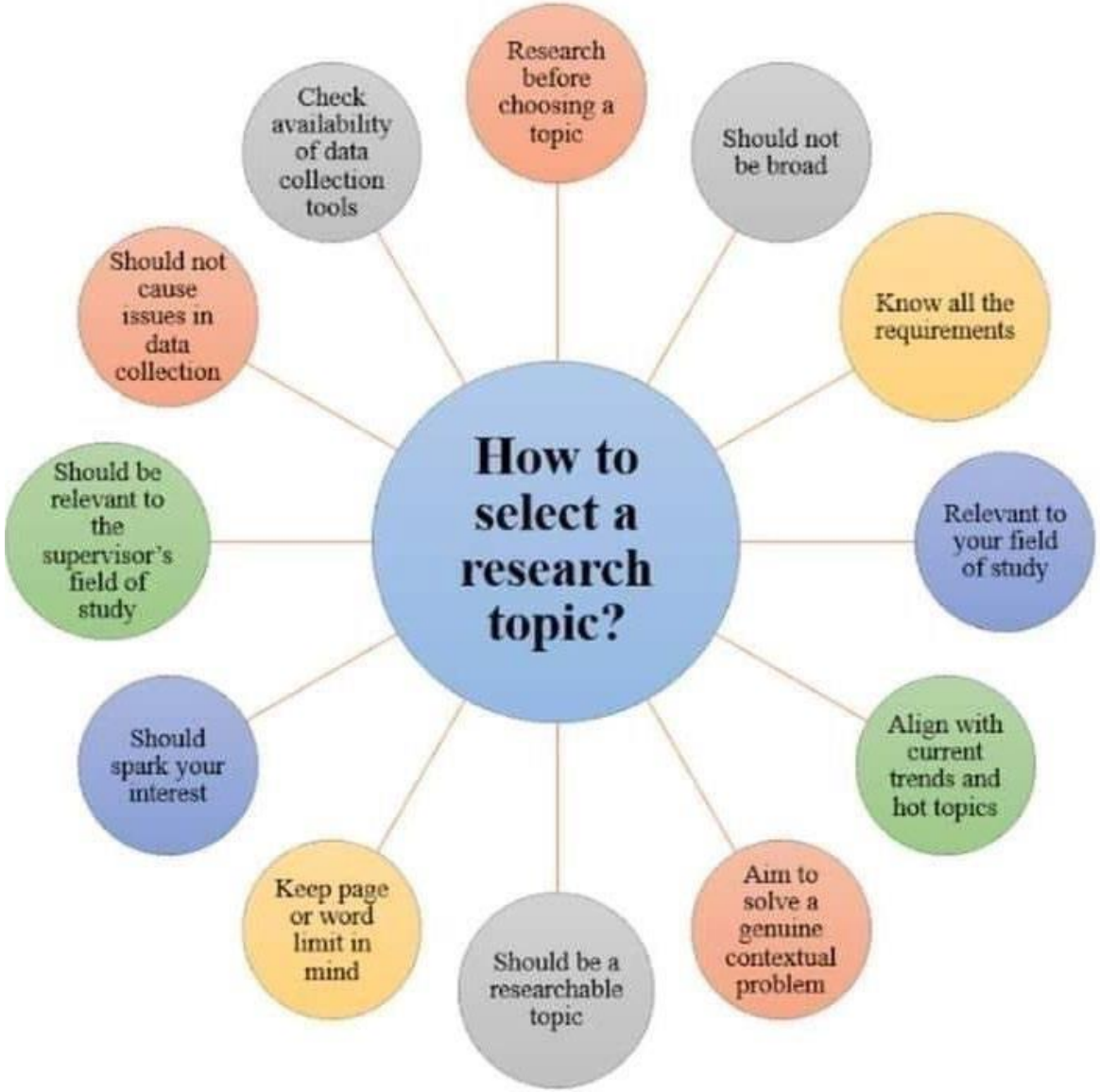
Agenda Today

- Choose a topic
- Locating good-quality journals and conferences
- Looking through the Proceedings
- Identify an area and topic - the structure of a conference (CFPs)
- Exercise: Reading and summarizing a research paper

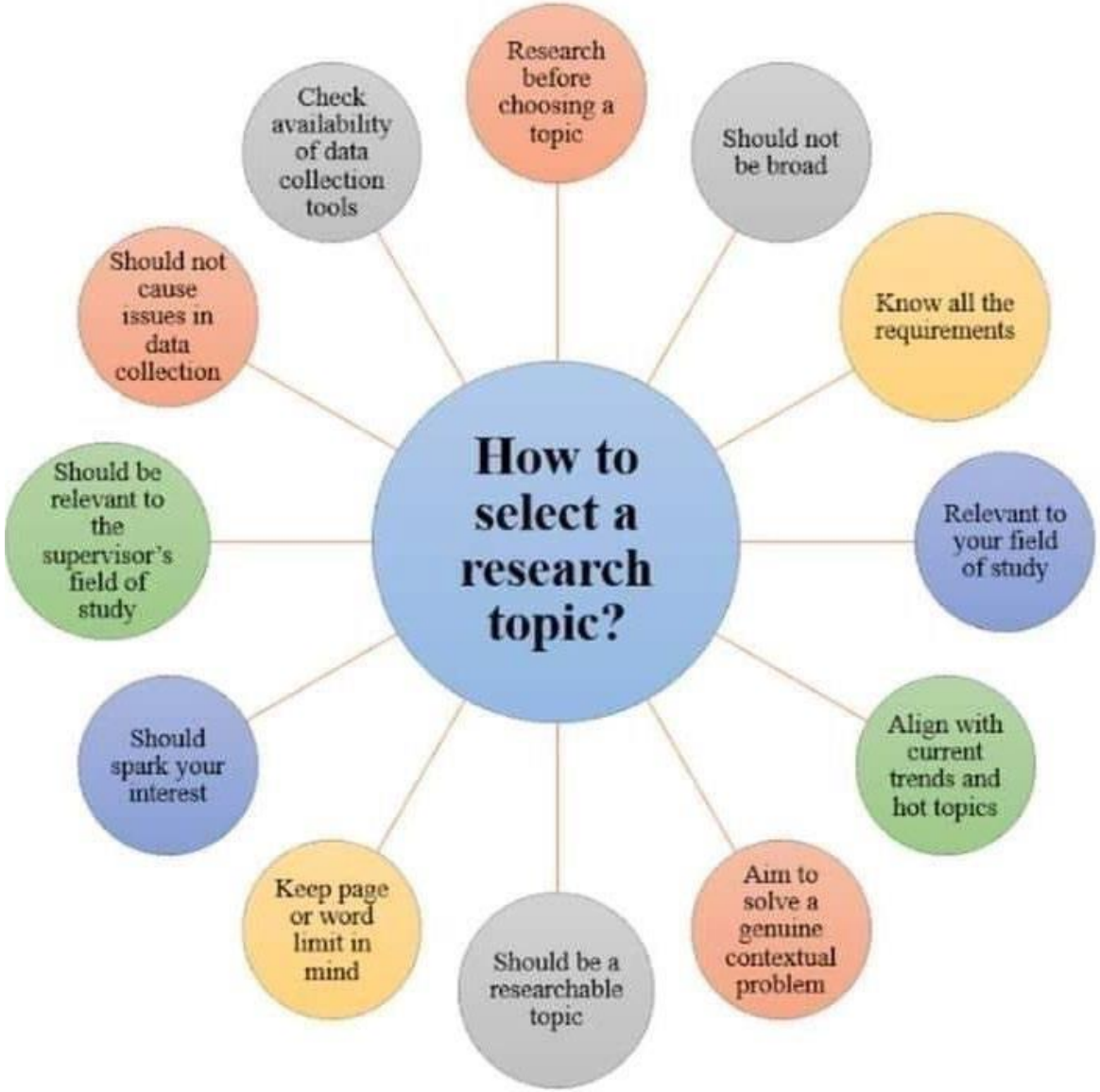
Recommendations on choosing your topic
- Know your strengths and interests



Recommendations on choosing your topic
- Read a lot, get inspired by the work of others



Recommendations on choosing your topic
- “Standing on the shoulders of giants”

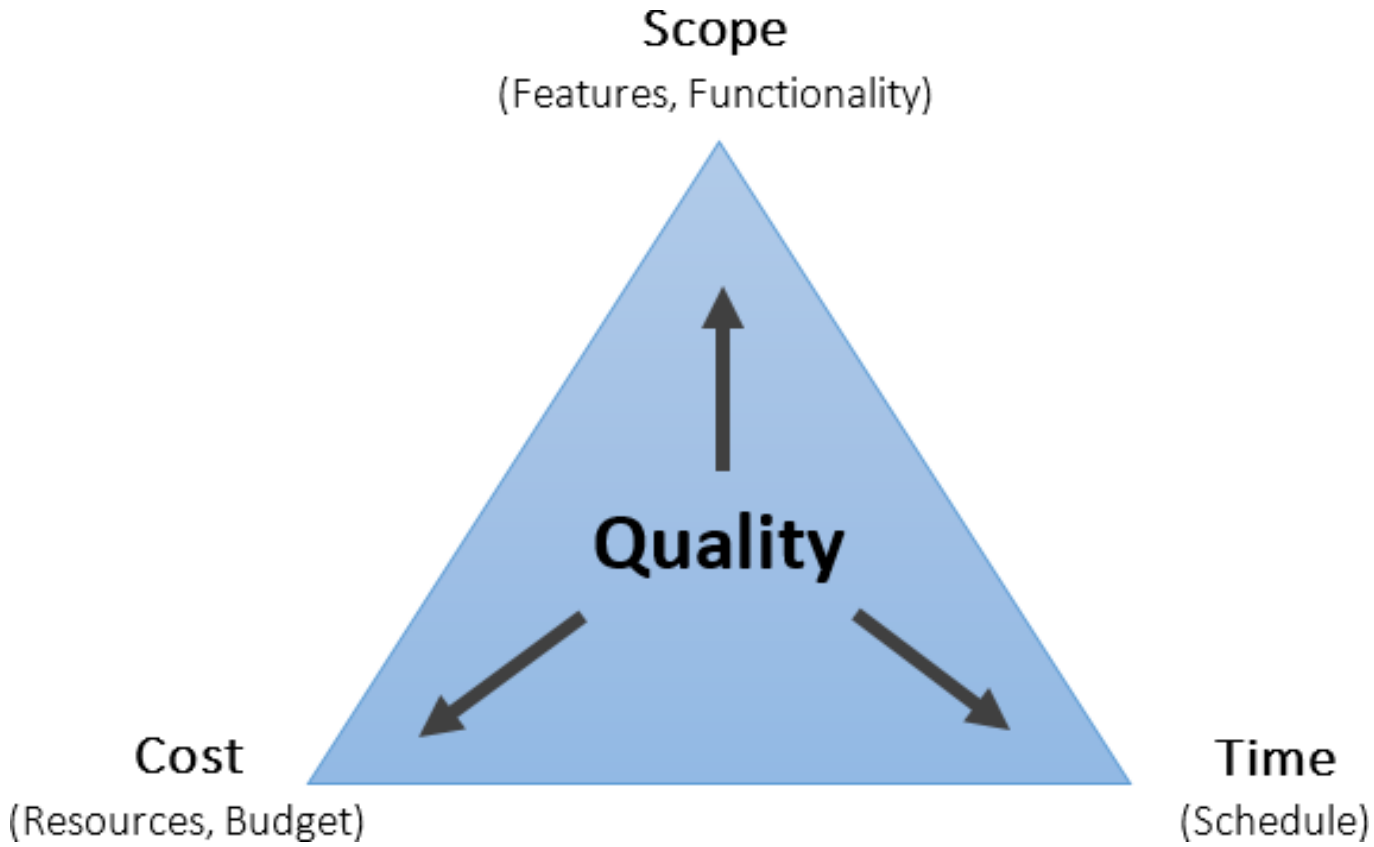


Recommendations on choosing your topic
- Be realistic about the scale of your research

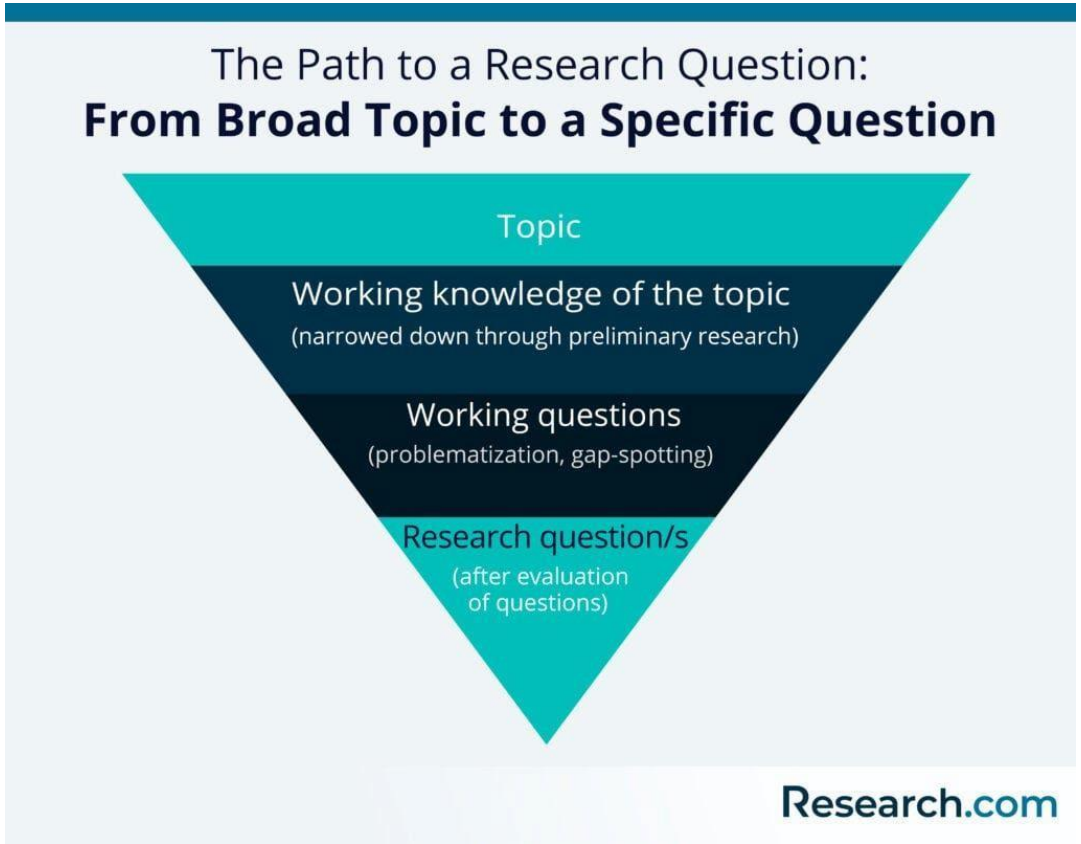
- **Specific**: Not a “fishing expedition”
- **Measurable**: Testable (statistically)
- **Attainable**: Something that “you” can do
  - You have the expertise
  - You have the equipment and resources
- **Realistic**:
  - Based on some sort of rational physics
  - Can be done in a reasonable time frame
- **Timely**:
  - Novel
  - Preferably based in new technologies (has a past).
  - Important to current concerns (has a present).
  - Can lead to new directions – i.e. is not an endpoint in itself (has a future).

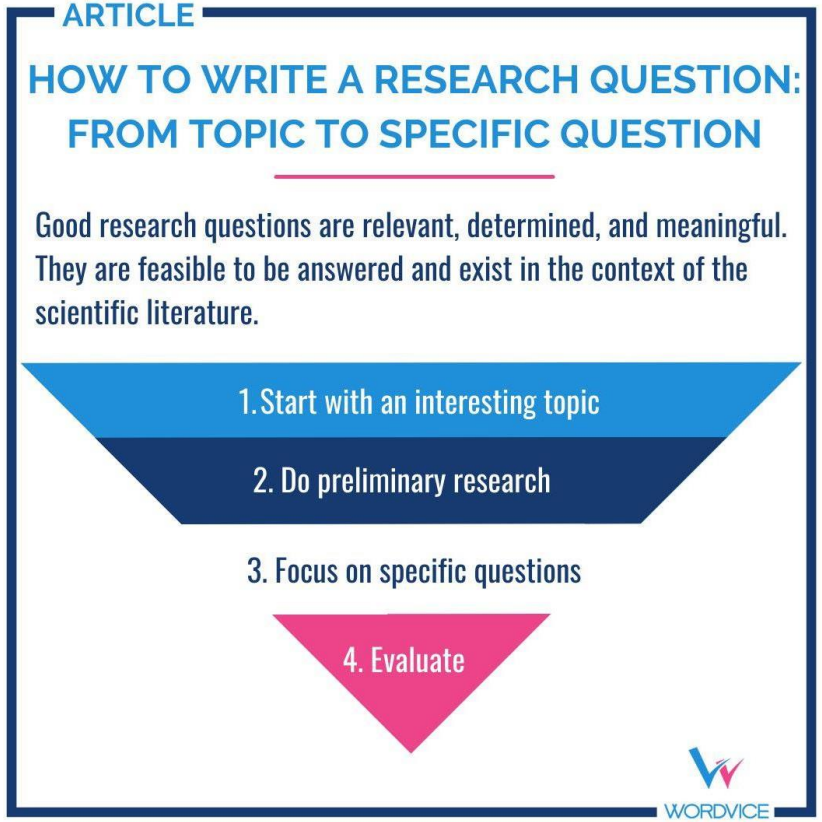


Recommendations on choosing your topic
The triangle of project management



Path to Research Question





Selecting a Research Problem in Computer Science

- Identify the area(s) that you want to work on
- Retrieve key terms that represent your research topic
- Look for ideas in the “Call for papers” from conferences
  - http://www.conferenceranks.com/
  - https://www.core.edu.au/conference-portal
- Once you have the topic, identify your research problem by doing a first round of literature review to find the gaps in what is currently know about it

Locate a Journal and a Conference

https://www.core.edu.au/conference-portal

Conferences are assigned to one of the following categories:

- A* - flagship conference, a leading venue in a discipline area
- A - excellent conference, and highly respected in a discipline area
- B - good conference, and well regarded in a discipline area
- C - other ranked conference venues that meet minimum standards
- Australasian - A conference for which the audience is primarily Australians and New Zealanders
- Unranked - A conference for which no ranking decision has been made
- National - A conference which is run primarily in a single country, with Chairs from that country, and which is not sufficiently well known to be ranked. (Papers and PC may be international)
- Regional - Similar to National but may cover a region crossing national borders.

Searching for Papers

- Proceedings (IEEE, ACM, Springer, Elsevier)
- DBLP (proceedings, authors)
- Scopus
- Google Scholar

Identify an Area and Topic

- **Journals**
  - IEEE Transactions on Cloud Computing (TCC)
  - ACM Transactions on Cloud Computing (TCC)
  - Future Generation Computer Systems (FGCS) (Elsevier)
  - Journal of Cloud Computing (Springer)
  - IEEE Internet Computing
- **Conferences**
  - USENIX Symposium on Networked Systems Design and Implementation (NSDI)
  - Usenix Network and Distributed System Security Symposium (NDSS)
  - ACM Symposium on Cloud Computing (SoCC)
  - USENIX Annual Technical Conference (ATC)
  - IEEE International Conference on Cloud Computing (CLOUD)
  - IEEE/ACM International Conference on Utility and Cloud Computing (UCC)

Example Research Areas for MSc Cloud Computing:
- Cloud resource allocation & autoscaling.
- Kubernetes scheduling and container orchestration.
- Serverless computing (FaaS) performance.
- Cloud cost optimisation and pricing models.
- Cloud security & isolation (VMs, containers, confidential computing).
- Cloud–edge–IoT integration.
- AI/ML workload optimisation on cloud platforms.
- Green cloud computing & energy efficiency.

Each of these topics has dozens of recent papers in SoCC, NSDI, ATC, IEEE CLOUD, FGCS, and IEEE TCC.

How to pick articles

- Pick **a recent survey** of the field so you can quickly gain an overview,
- Pick a paper that you can **easily understand** – book chapters often give easier understandable materials and lengthy explanation that may give you a head start although they may not be as up-to-date as papers
- Pick **papers that are related to each other** in some ways and/or that are in the same field so that you can write a meaningful survey out of them
- Favour papers from **well-known journals and conferences**
- Favour **"foundational"** papers in the field (as indicated in other people's survey paper),
- Favour more **recent papers**
- Find relationships with respect to each other and to your topic area (classification scheme/categorization)

How to pick articles

- How to tell that a paper is relevant to your topic?
  - Skim the paper (title, abstract, conclusions)
- Document your literature review
  - For each relevant paper, include your own commentary on the significance of the approach and the solutions presented in each paper
  - Provide a critical assessment of the work that has been done

Exercise: Locate a Journal and a Conference

1. Identify 2 journals/articles and 2 conferences papers **related** to your research topic that are published in the past 5 years. Tip: Review the title and the abstract to decide which papers will be relevant

2. For the journals locate JCR impact factor, for the conferences locate the conference ranking (https://www.core.edu.au/conference-portal), and quote it in your work.

3. Summarize the papers in a 1-2 paragraphs. The **summaries should be paraphrasing rather than copy-paste.**

4. Reference the paper in APA style, e.g.,

    1. Ayala-Rivera, V., McDonagh, P., Cerqueus, T., & Murphy, L. (2014). A systematic comparison and evaluation of k-anonymization algorithms for practitioners. *Transactions on data privacy*, *7*(3), 337-370.

.